Blank Template for LING 581 Research Paper–Your Title Goes Here

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Abstract

A brief summary of the research, including the purpose, methods, results, and conclusions.

[For your final paper submission, you may use an AI-generated summary of your manuscript for the Abstract section only so long as you cite the proper AI attribution like this (OpenAI, 2024).]

1 Introduction

Introduction to the topic, background information, research questions, and the significance of the study.

2 Literature Review

Review of relevant previous research and theoretical framework.

3 Methodology

Detailed description of the research design, data collection, and analysis methods.

4 Results

Presentation of the findings, including tables, figures, and statistical analyses.
Instagram Comments (test) - 56 samples

Model	Accuracy	Latency (s)
Gemma 4 E2B	60.71%	0.0930
Qwen3.5 0.8B Q4	28.57%	0.2128
Qwen3.5 2B Q4	28.57%	0.5344
BERTweet Sentiment	60.71%	0.0081
Reviews-SST2	46.43%	0.0035
Twitter-RoBERTa	60.71%	0.0062

Table 1: Benchmark results for Instagram Comments (test). Accuracy represents the maximum accuracy achieved across multiple runs; latency is the average inference time per sample.

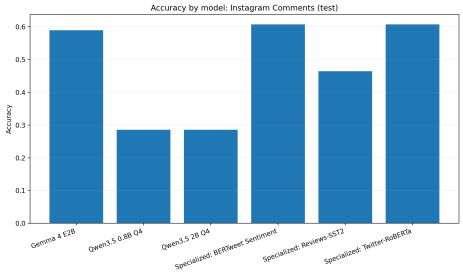

Figure 1: Accuracy chart for Instagram Comments (test).

Kaggle Reviews (dolbokostya) - 2000 samples

Model	Accuracy	Latency (s)
Gemma 4 E2B	63.00%	0.0851
Qwen3.5 0.8B Q4	65.00%	0.1559
Qwen3.5 2B Q4	65.00%	0.3159
BERTweet Sentiment	63.00%	0.0060
Reviews-SST2	68.00%	0.0031
Twitter-RoBERTa	65.00%	0.0060

Table 2: Benchmark results for Kaggle Reviews (dolbokostya). Accuracy represents the maximum accuracy achieved across multiple runs; latency is the average inference time per sample.

Kaggle Sentiment Analysis (mdismielhossenabir) - 499 samples

5 Discussion

Interpretation of the results, implications, and relation to previous research.

6 Conclusion

Summary of the main findings, limitations of the study, and suggestions for future research.

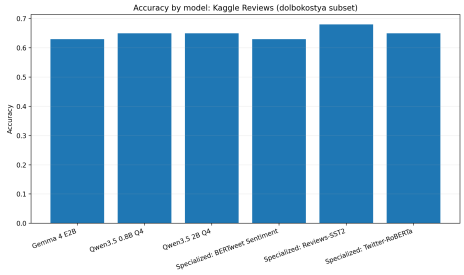

Figure 2: Accuracy chart for Kaggle Reviews (dolbokostya).

Model	Accuracy	Latency (s)
Gemma 4 E2B	73.21%	0.0976
Qwen3.5 0.8B Q4	33.87%	0.2421
Qwen3.5 2B Q4	32.26%	0.4828
BERTweet Sentiment	71.94%	0.0059
Reviews-SST2	50.70%	0.0033
Twitter-RoBERTa	71.94%	0.0061

Table 3: Benchmark results for Kaggle Sentiment Analysis (mdismielhossenabir). Accuracy represents the maximum accuracy achieved across multiple runs; latency is the average inference time per sample.

References

OpenAI. 2024. ChatGPT: A Large-Scale Generative Model for Open-Domain Chat. https://github.com/openai/gpt-3.

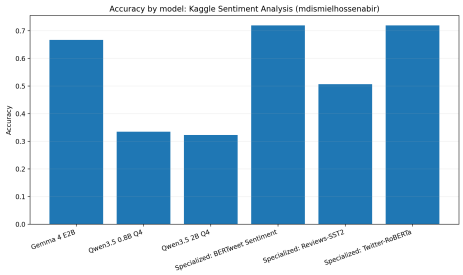

Figure 3: Accuracy chart for Kaggle Sentiment Analysis (mdismielhossenabir).